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INSTRUMENTAL-VARIABLE ESTIMATION OF COUNT DATA MODELS: APPLICATIONS TO MODELS OF CIGARETTE SMOKING BEHAVIOR

John Mullahy*

Abstract—As with most analyses involving microdata, applications of count data models must somehow account for unobserved heterogeneity. The count model literature has generally assumed that unobservables and observed covariates are statistically independent. Yet for many applications this independence assumption is clearly tenuous. When the unobservables are omitted variables correlated with included regressors, standard estimation methods will generally be inconsistent. Though alternative consistent estimators may exist in special circumstances, it is suggested here that a nonlinear instrumental-variable strategy offers a reasonably general solution to such estimation problems. This approach is applied in two examples that focus on cigarette smoking behavior.

I. Introduction

OVER THE past decade, models for count data have become increasingly popular in applied econometrics. Confined no longer to the analysis of rare events, models such as the Poisson regression model have been employed to analyze a variety of problems where the dependent variable y is measured as a nonnegative integer.¹

As is the case when any statistical model is applied to microdata, applications of count models must recognize and account appropriately for the likely presence of unobserved heterogeneity in the data-generating process. To date, the count model literature has focused almost exclusively on the case where the unobserved heterogeneity Θ and the measured regressors x are statistically independent, that is, $h(\Theta|x) = h(\Theta)$. The leading example considered in the literature is that where the baseline model is Poisson and the heterogeneity is gamma distributed, in which case the negative binomial model results as a convenient closed-form gamma mixture of Poissons.

Yet for many cross-sectional applications such as independence assumption will be tenuous at best. When instead the unobserved heterogeneity is appropriately characterized as “unmeasured regressors correlated with x ,” then maximum likelihood (ML), quasi-ML (QML), nonlinear least squares (NLLS), and other standard estimators that are consistent when statistical independence of x and Θ obtains will not generally be consistent when independence fails.² Inconsistency results in such cases because the typical term in the

estimating equations defining the standard estimators, say, $q(x_i)[y_i - \exp(x_i\alpha)]$, does not have zero expectation at the true model so that the estimating equations defining such estimators, $\sum_{i=1}^N q(x_i)[y_i - \exp(x_i\alpha)] = 0$, will have biased and inconsistent solutions.

Consider the following illustrative example, suggested by the work of Cameron and Trivedi (1986). Suppose the demand for some form of medical care y is measured as a count—such as the number of physician visits or consultations during a two-week interval—and estimation via Poisson or related regression methods is contemplated. The demand function might be described as $y = f(s, v, \Theta; \alpha)$, where s is a measure of insurance generosity, v is a vector of exogenous covariates, and Θ are determinants of demand unobserved by the econometrician (e.g., illness propensities of household members). A standard result in health economics is that s and Θ are unlikely to be uncorrelated in the population since insurance generosity can—at least in some countries—be self-selected according to a household's ex ante anticipation of its demand for care. Failure to account for the nonzero correlation between s and Θ would in general result in inconsistent estimates of the parameters of interest, α .

This paper suggests a general nonlinear instrumental-variable (IV) strategy for consistent estimation in such instances.³ The critical issue in developing such an estimator turns out to be how Θ enters the model. In the existing count model literature that treats x and Θ as statistically independent, it is generally assumed that Θ and x play “symmetric” roles in the data-generating process. That is, in generalized linear model parlance, the “linear predictor” is $x\alpha + \Theta$. If so, then (as shown below) a seemingly obvious nonlinear IV estimator will not generally consistently estimate α when x and Θ are correlated. This is true even if otherwise seemingly reasonable instrumental variables z are utilized since the fundamental orthogonality requirement between instruments and residual functions is violated, the crux of the problem being that a standard assumption in nonlinear IV estimation—that unobservables be additively separable from the parametric structural model—does not generally hold.⁴

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¹ The recent survey by Gurmu and Trivedi (1994) provides an excellent overview of event count models.

² This statement is actually a bit strong. It is shown below that a weaker mean-independent condition is generally sufficient for consistency to be obtained.

³ Alternative consistent estimators that do not rely on the availability of “classical” instrumental variables may be available in special circumstances. For example, Hausman et al. (1984) and Wooldridge (1994) have proposed estimators that are applicable when panel data are available.

⁴ See Manski (1988, secs. 2.5 and 6.1) on the role of additive separability in IV estimation. As indicated below, IV methods have been proposed wherein consistency will generally require that Θ enter the structural model in an awkward, “nonsymmetrical” manner (i.e., $y = \exp(x\alpha) + \Theta + \epsilon$) that does not follow naturally from treating Θ as unobserved regressors in a Poisson probability model. Mullahy and Portney (1990) circumvented this issue by assuming a linear specification, $y = x\alpha + \Theta + \epsilon$, for their application; while linear count model specifications cannot be ruled out a priori, they can pose problems for estimation and prediction.

It is demonstrated, however, that a particular transformation of the model eliminates these problems. A transformed residual function is defined such that the key orthogonality condition will generally be obtained for suitable z . Consistency and asymptotic normality follow when estimation of the transformed model is undertaken in a generalized method of moments (GMM) framework. In particular, consistency of the proposed estimator is obtained under standard nonlinear IV assumptions on the relationships between instruments and unobservables (e.g., mean independence).⁵

The plan for the paper is as follows. The background and preliminaries are laid out in section II. Section III describes consistent and inconsistent IV estimation. Section IV considers the merits of some alternative approaches. Section V applies the proposed method to models of cigarette smoking behavior and birthweight determinants. Section VI offers some concluding remarks.

II. Preliminaries

A. Count Data Regression Models

Let y be the count-valued outcome variable ($y \in \{0, 1, 2, \dots\}$) of interest with

$$\begin{aligned} E(y|x, \eta; \alpha) &= \exp(x\alpha)\eta \\ &= \exp(\alpha_0 + x_1\alpha_1)\eta \end{aligned} \quad (1)$$

where $x = [1, x_1]$ is the $1 \times k$ regressor vector, $\alpha = [\alpha_0, \alpha_1']'$ is the unknown $k \times 1$ parameter vector,⁶ and $\eta = \exp \Theta > 0$ is unobserved and varies over the population with marginal distribution $H(\eta)$. Since a constant term is assumed to be included in x , then $E(\eta) = 1$ can be assumed without loss of generality. Leading cases of this exponential mean specification are the Poisson model and the geometric model.

The observable data are N independent observations on $D_i = (y_i, x_i, z_i)$, where z_i is a vector of instrumental variables (whose properties are described below) that may share elements in common with x .⁷ Note that this specification treats $x\alpha$ and Θ symmetrically in the manner described in the introduction. The complication is that Θ may be a “problematic” unobservable in the sense of being correlated with some subvector of x .

⁵ It should be noted at the outset that since the standard count data regression model is essentially an exponential regression model, then the restriction to integer-valued y is itself largely inessential. Thus for purposes of this paper “count data models” can be conceived more broadly as exponential regression models for nonnegative dependent variables. The basic ideas apply to any model for nonnegative dependent variables having exponential conditional means of the particular form described below.

⁶ To minimize notational clutter, no distinction will be made between a generic value of the parameter vector α and its true value.

⁷ Subscripts indexing observations will be suppressed throughout unless needed for clarity.

Assumption (1) implies the exponential regression model

$$\begin{aligned} y &= E(y|x, \eta; \alpha) + \epsilon \\ &= \exp(x\alpha)\eta + \epsilon \end{aligned} \quad (2)$$

where $E(\epsilon|x, \eta) = 0$ by construction.⁸

Equation (2) will henceforth be the main focus of the paper. Indeed, it is useful to regard equation (2) as defining the data-generating process for y , y being determined as $\exp(x\alpha)\eta + \epsilon$ given realizations of (x, η, ϵ) with ϵ mean-independent of x and η .⁹ Whether equation (2) is obtained from a Poisson, a negative binomial, or another underlying model is largely irrelevant. The key attribute of equation (2) is that y is a nonnegative, though not necessarily a count, variable.¹⁰

B. Inconsistency of Standard Estimators

So long as $E(\eta|x) = E(\eta)$ —a special case being statistical independence of η and x —then a standard result from QML theory is that the QML estimator consistently estimates α_1 (see Gourieroux et al. (1984) and Wooldridge (1997)). The usual assumption in the count data literature is that η and x are statistically independent, but the weaker condition in fact suffices for consistency.

Conversely, if $E(\eta|x) \neq E(\eta)$, QML estimators as well as alternative related estimators (e.g., NLLS) will generally not estimate α_1 consistently. As such, if Θ has the character of omitted regressors correlated with x , then estimators commonly used in count data model applications will not generally be consistent for α_1 . In a nutshell, this results because of failure of the key population orthogonality condition required for consistency,

$$\begin{aligned} E[q(x)'(y - \exp(x\alpha))] &= E[q(x)'(\exp(x\alpha)(\eta - 1) + \epsilon)] \\ &= E_x[q(x)' \exp(x\alpha)E[(\eta - 1)|x]] + E[q(x)'\epsilon] \\ &= E_x[q(x)' \exp(x\alpha)E[(\eta - 1)|x]] + 0 \\ &\neq 0, \text{ in general} \end{aligned}$$

where $q(x)$ is the vector of “instruments” implied by the particular estimator [$q(x) = x$ for Poisson QML; $q(x) =$

⁸ Note that ϵ appears in equation (2) as an additive “regression error.” As Wooldridge (1992) has recently pointed out, the “regression error” in such a model could also be multiplicative as the two will be observationally indistinguishable. The results are carried out here in the more general context of the additive error formulation since, were the regression error multiplicative, it could be absorbed as part of η .

⁹ Note that α_0 and $E(\eta)$ are essentially nuisance parameters that are not separately identified. In what follows, “consistent estimation of α ” is shorthand for “consistent estimation of $\alpha_0 + \ln(E(\eta))$ and of α_1 .” Under the standard normalization $E(\eta) = 1$, the distinction is unnecessary.

¹⁰ A somewhat more general model that would also be amenable to estimation by the methods proposed here is $y = g(x; \alpha)\eta + \epsilon$, $g(x; \alpha) > 0$. Symmetry of $x\alpha$ and Θ as well as guaranteed positivity of $g(x; \alpha)$ for any x and α suggest, however, that restricting attention to the particular case $g(x; \alpha) = \exp(x\alpha)$ is not unduly restrictive for many applications.

$\exp(x\alpha)$ x for NLLS; etc.). That $E[(\eta - 1)|x] \neq 0$ is the essence of the inconsistency.

Using all elements of x to form the instruments $q(x)$ in the estimating equations is thus unlikely to yield consistent estimates of α when the mean-independence assumption fails. Given the availability of an appropriate set of instrumental variables z , one might hope to overcome this hurdle. As the results in the next section show, consistent estimation of α is indeed generally possible given a suitably defined residual function and given instrumental variables z that satisfy the standard requirements for instrumental variables in nonlinear IV estimation.

III. Instrumental-Variable Estimation

A. Basic Issues

The ideas here derive fundamentally from the literature on nonlinear IV or GMM estimators (see Amemiya (1974, 1985), Hansen (1982), and Manski (1988)), in conjunction with the recognition that standard consistency results for such estimators are derived by postulating structural models having unobservable components that are, in a particular sense, additively separable.¹¹ These methods require instrumental variables z having certain statistical relationships with x , η , and ϵ , amounting to the standard requirements for instrumental variables in nonlinear IV estimators (e.g., Amemiya (1985, ch. 8)). The trick is to transform the model into a form that satisfies standard conditions for consistency of such IV estimators.

Let z be a p -vector ($p \geq k$) of candidate instrumental variables that may contain those elements of x with which η is mean independent, and let the joint distribution of (x, z, η, ϵ) be denoted by $F(x, z, \eta, \epsilon)$. The key assumptions on the instrumental variables z are that $E(y|x, \eta, z) = E(y|x, \eta)$ and that $E(\eta|z) = \tau$, where τ is a constant not depending on z . As described below, the normalization $\tau = 1$ can again be invoked so long as x contains a constant term.

Recall from equation (2) that $E(\epsilon|x, \eta) = 0$ is implied by the model's regression structure, as ϵ is by definition simply the difference between a random variable y and its expectation conditional on (x, η) . For present purposes, however, somewhat stronger assumptions are required. It is assumed henceforth that $E(\epsilon|x, z, \eta) = 0$. While stronger than the restriction implied by the regression model itself, it would in any event appear natural to require candidate instrumental variables to satisfy such restrictions.

Consistency of nonlinear IV estimators is typically established on the basis of a residual function $u(y, x; \alpha) = u(\epsilon, x, \eta; \alpha)$ satisfying the conditional moment restriction $E[u(y, x; \alpha)|z] = 0$, which in turn implies the zero correlation

condition

$$E[\pi(z)'u(y, x; \alpha)] = E[m(y, x, z; \alpha)] = 0$$

where $\pi(z)$ is an s -vector ($s \geq k$) of functions of z , and where the existence of all expectations is assumed. The sample analog of this population zero correlation, $m_N(\alpha) = N^{-1} \sum_{i=1}^N \pi(z_i)'u(y_i, x_i; \alpha)$, is typically used as the basis of the quadratic nonlinear IV estimand $m_N(\alpha)'Wm_N(\alpha)$, where W is some positive definite matrix.¹²

B. An Inconsistent IV Estimator

What might seem an obvious approach to IV estimation of α , henceforth termed IV-A, would be to specify the residual function

$$\begin{aligned} u_A(y, x; \alpha) &= y - \exp(x\alpha) \\ &= \exp(x\alpha)(\eta - 1) + \epsilon \end{aligned}$$

where the second equality follows from equation (2).¹³ However, for this specification of the residual function $E[u_A(y, x; \alpha)|z] = 0$ will not in general hold if equation (2) is the true model. Nonlinear IV estimators based on the residual function $u_A(y, x; \alpha)$ will thus not generally be consistent for α , this being due essentially to the fact that η in the true model (2) is not additively separable from x in the residual function $u_A(y, x; \alpha)$. Using iterated expectations, this is seen as follows:

$$\begin{aligned} E[u_A(y, x; \alpha)|z] &= E[y - \exp(x\alpha)|z] \\ &= E[\exp(x\alpha)(\eta - 1)|z] + E[\epsilon|z] \\ &= E_x[\exp(x\alpha)E\{(\eta - 1)|x, z\}|z] + 0 \\ &= E_x[\exp(x\alpha)\phi(x, z)|z] + 0 \\ &\neq 0, \text{ in general} \end{aligned}$$

where $\phi(x, z) = E[(\eta - 1)|x, z]$ is in general nonzero. The essence of the problem is that functions of x and η are multiplied together in $u_A(y, x; \alpha)$.

Should η enter the structural model additively, that is, if instead of equation (2) one maintained

$$y = \exp(x\alpha) + \eta + \epsilon \quad (3)$$

then the IV-A approach would in general be consistent given suitable instrumental variables z . Equation (3) is precisely the form—additively separable in unobservables—for which

¹¹ That is, the standard formulation of nonlinear structural models, $u_0(z, b) - u = 0$, specifies the unobservables u as additively separable from functions $u_0(\cdot)$ of observables z . See Manski (1988, secs. 2.5 and 6.1). This is Manski's notation, where z summarizes all observable data and b is the parameter vector to be estimated.

¹² The main identification assumption for such estimators is full rank of the derivative matrix $\text{plim}[N^{-1} \sum_{i=1}^N \partial m(y_i, x_i, z_i; \alpha)/\partial \alpha]$. This is "strong asymptotic identifiability," in the sense of Davidson and MacKinnon (1993, theorem 5.2 and p. 594); see also Amemiya (1985, theorem 8.1.1). Identification in this sense will be assumed henceforth.

¹³ An analogous approach was suggested by Grogger (1990), whose concern was more with testing for correlation between x and Θ than with consistent estimation per se.

Amemiya (1974) originally established consistency of non-linear IV estimators. A specification such as equation (3), in which η enters additively, is awkward to motivate directly, however, as it fails to treat observables and unobservables symmetrically as determinants of y .

C. Consistent IV Estimation

The central result of this paper is that a transformation of the regression model (2) averts the inconsistency of estimators based on the IV-A approach. This is accomplished by transforming equation (2) to obtain a residual function that is additively separable in the problematic unobservables η . This approach will be referred to as IV-T (for “transformation”).

Recall that the key assumptions on the instrumental variables z are that $E(\epsilon|x, z) = 0$ and that $E(\eta|z)$ is a constant not depending on z , normalized to equal 1. Transforming the regression model by dividing both sides of equation (2) by $\exp(x\alpha)$ gives

$$\exp(-x\alpha)y = \eta + \exp(-x\alpha)\epsilon. \quad (4)$$

Define $T(y, x; \alpha) = \exp(-x\alpha)y$ and $v = \eta - 1$, so that equation (4) can be rewritten as

$$T(y, x; \alpha) - 1 = v + \exp(-x\alpha)\epsilon. \quad (5)$$

Estimators based on IV-T use $T(y, x; \alpha) - 1$ as the residual function. Note that the transformation has resulted in an additively separable unobservable, $v = \eta - 1$, in the sense described earlier.

Consistency of IV-T estimators is established by first demonstrating the validity of the conditional moment restriction $E[T(y, x; \alpha) - 1 | z] = 0$, which follows given the previous assumptions on the instrumental variables z ,

$$\begin{aligned} E[T(y, x; \alpha) - 1 | z] &= E[\exp(-x\alpha)\epsilon + v | z] \\ &= E[\exp(-x\alpha)\epsilon | z] + E(v | z) \\ &= 0. \end{aligned}$$

This follows since the first term is zero by iterated expectations and the second term is zero since $E(\eta|z) = 1$. In contrast to the IV-A approach, where functions of x and η are multiplied in the residual function, the key feature of IV-T is that the transformation results in a residual function in which the problematic unobservable is made additively separable from the endogenous variables.¹⁴

D. Identification, Estimation, and Inference

If $E(\eta|z) = \tau$ were a free parameter, then multiplication of the left-hand side of equation (5) by any nonzero constant ρ would give a valid conditional moment restriction, that is,

¹⁴ See Wooldridge (1994) for a discussion of an approach analogous to IV-T for applications in panel data settings.

$E[\rho(T(y, x; \alpha) - \tau)|z] = 0$. As in the standard count model literature, τ and α_0 are not separately identified; the normalization $\tau = 1$ is thus used.

IV-T can be operationalized as a GMM estimator¹⁵ that minimizes a criterion function of the form

$$\begin{aligned} \Gamma_N(\alpha) &= \left[N^{-1} \sum_{i=1}^N m(y_i, x_i, z_i; \alpha) \right]' \\ &\quad \times \Omega^{-1} \left[N^{-1} \sum_{i=1}^N m(y_i, x_i, z_i; \alpha) \right] \end{aligned}$$

where Ω^{-1} is an $s \times s$ positive definite weighting matrix, and

$$m(y_i, x_i, z_i; \alpha) = \pi(z_i)' u_T(y_i, x_i; \alpha), \quad s \times k.$$

Given the instrument vector $\pi(z)$, the optimal GMM estimator $\hat{\alpha}_{\text{GMM}}$ sets

$$\Omega = \text{var} \left[\frac{\sum_{i=1}^N m(y_i, x_i, z_i; \alpha)}{\sqrt{N}} \right]$$

and uses in applications an estimate of Ω ,

$$\hat{\Omega} = N^{-1} \sum_{i=1}^N m(y_i, x_i, z_i; \hat{\alpha}) m(y_i, x_i, z_i; \hat{\alpha})'$$

where $\hat{\alpha}$ is some preliminary consistent estimate (e.g., nonlinear two-stage least squares).

Assuming regularity conditions are satisfied,¹⁶ then inferences can be conducted by treating $\sqrt{N}(\hat{\alpha}_{\text{GMM}} - \alpha)$ as asymptotically normal with mean zero and asymptotic covariance matrix given by

$$\text{var} [\sqrt{N}(\hat{\alpha}_{\text{GMM}} - \alpha)] = [G(\alpha)' \Omega^{-1} G(\alpha)]^{-1}$$

with

$$\begin{aligned} G(\alpha) &= \text{plim} \left[\frac{N^{-1} \sum_{i=1}^N \partial m(y_i, x_i, z_i; \alpha)}{\partial \alpha} \right] \\ &= \text{plim} \left[N^{-1} \sum_{i=1}^N -\pi(z_i)' T(y_i, x_i; \alpha) x \right], \quad N \times k. \end{aligned}$$

¹⁵ See Davidson and MacKinnon (1993, ch. 17) for a thorough treatment of GMM estimation and inference.

¹⁶ Consistency and asymptotic normality follow from standard regularity conditions spelled out by Davidson and MacKinnon (1993, sec. 17.2). Most fundamentally, the population version of the objective function $\Gamma(\alpha)$ must be uniquely minimized at the true value of α for consistency to be obtained.

The statistic $N\Gamma_N(\hat{\alpha}_{\text{GMM}})$ is distributed $\chi^2_{(s-k)}$ under the null hypothesis of orthogonality between instruments and residuals. As such, this statistic (sometimes referred to as Hansen's J -statistic) serves as a test of the validity of overidentifying restrictions if any are present (i.e., if $s > k$). In addition, a consistent estimator like IV-T can serve as the basis of a Hausman-type test where, for example, the IV-T estimates could be compared with the ML estimates obtained under standard assumptions.

E. An Analogy

To better appreciate how the IV-T estimator works, an analogy may be useful. For strictly positive y , the standard has long been the log-linear specification

$$y = \exp(x\beta)\omega.$$

For this model, if it is suspected that regressors x are correlated with unobservables ω , then the standard (and most straightforward) approach to consistent estimation of β is to assume that $E[\ln(\omega)|z]$ is a constant not depending on z , and then to apply a linear IV estimator to the log-transformed model

$$\ln y = x\beta + \ln \omega$$

a formulation that satisfies additive separability. Were it assumed, alternatively, that $E[\omega|z]$ is a constant not depending on z (as above), then the slope parameters in β in this model could be consistently estimated using the nonlinear IV-T method by transforming and using the residual function $u_T(y, x; \beta) = T(y, x; \beta) - 1$. As such, under a somewhat stronger assumption than that used above—namely, that ω and z are statistically independent¹⁷—both the linear IV estimator applied to the log-transformed model and the IV-T estimator applied to the residual function $u_T(y, x; \beta)$ should be consistent. A major discrepancy might suggest that some form of misspecification is present.

IV. Alternative Approaches

A. Likelihood-Based Methods

A first alternative to the IV-T method is available if the functional form of the conditional probability $f(y|x)$ is known up to certain parameters, in which case consistent estimates can be obtained via likelihood-based methods. If $f(y|x, \eta; \alpha)$ is assumed known up to parameters α (e.g., y is Poisson distributed with parameter $\lambda(x, \eta; \alpha)$), and if $h(\eta|x; \gamma)$ can be assumed known up to parameters γ (e.g., η is exponentially distributed with parameter $\zeta(x; \gamma)$), $f(y|x)$

can be obtained as

$$f(y|x; \alpha, \gamma) = \int f(y|x, \eta; \alpha) dH(\eta|x; \gamma).$$

Having integrated out the observables, then (α, γ) —or, more precisely, the identified elements of (α, γ) —can be estimated by ML applied to the likelihood function built upon the conditional distributions $f(y|x; \alpha, \gamma)$.¹⁸ This approach is fully analogous to the method proposed by Chamberlain (1980) for estimating probit models where random effects are correlated with observables.

While the appeal of such a method is obvious given knowledge (up to parameters) of the relevant conditional probabilities, maintaining reasonable assumptions on the structures of $h(\eta|x; \gamma)$ (in particular) and of $f(y|x, \eta; \alpha)$ (to a perhaps lesser extent) may be tenuous. An approach less demanding of distributional assumptions would in most instances be preferable.

B. Two-Stage Methods

A second alternative is to invoke the results on ML or QML estimation of count data models in conjunction with an assumed knowledge (up to parameters) of a reduced-form model for x given z . This approach is referred to as the two-stage QML (2SQML) method.¹⁹ Suppose the reduced form for x is

$$\begin{aligned} x &= zR\Delta + v \\ &= z\delta + v \end{aligned}$$

where R is a $p \times p$ matrix that defines a priori zero restrictions on the $p \times k$ parameter matrix Δ as required for identification, that is, $\text{rank}(\delta) = k$. Note that z will typically contain some elements of x itself, in which case the corresponding row of R will contain $p - 1$ zeros and one unit element, with the corresponding elements of Δ and v obviously 1 and 0. To keep the analysis uncluttered, however, these elements of Δ and v can be treated as free parameters and unrestricted errors, respectively.²⁰

Assume that $\exp(v\zeta)\eta$ is mean independent of z for any finite vector of constants c . Substituting the reduced form for x into $\lambda(x, \eta; \alpha)$ gives

$$\begin{aligned} \lambda(x, \Theta; \alpha) &= \exp(x\alpha)\eta \\ &= \exp[(z\delta)\alpha + v\alpha]\eta \\ &= \exp(z\theta + \xi) \\ &= \lambda(z, \xi; \theta) \end{aligned}$$

¹⁸ Whether all elements of α and γ are identified is a separate issue altogether.

¹⁹ See Wooldridge (1997) for an alternative two-stage approach.

²⁰ The assumption of linearity of $E(x|z)$ in z is made here for simplicity. Newey (1985) points out in a related context that linearity of $E(x|z)$ in z is not necessarily required for consistency of this class of two-stage estimators.

¹⁷ Under this assumption, $E[g(\omega)|z]$ will be a constant not depending on z for any function $g(\cdot)$.

where $\theta = \delta\alpha$ ($p \times 1$) and $\xi = \nu\alpha + \ln \eta$. Since $\exp \xi$ is mean independent of z by assumption, a (quasi-) ML Poisson estimator based on $\lambda(z, \xi; \theta)$ is consistent for θ based on earlier arguments.

The ultimate aim, of course, is recovering a consistent estimate of α . This can be accomplished in several ways. Perhaps most straightforward is to obtain a consistent estimate of δ (e.g., ordinary least squares (OLS)), predict $z\delta$ as $\hat{x} = z\hat{\delta}$, and replace x with $\hat{x}$ in (quasi-) ML estimation. The resulting estimate of α is consistent, although the covariance matrix estimate would have to be adjusted to account for the sampling variation introduced from the estimation of δ .

In general, a more efficient estimator is available from the application of a minimum-distance (MD) estimator (see Amemiya (1985, p. 393), Lee (1981), and Newey (1985)). Using $\theta = \delta\alpha$, this is based on the relationship

$$\begin{aligned}\hat{\theta} - \hat{\delta}\alpha &= \theta - \delta\alpha + [(\hat{\theta} - \theta) - (\hat{\delta} - \delta)\alpha] \\ &= \theta - \delta\alpha + u\end{aligned}$$

where $\hat{\theta}$ is the (quasi-) ML estimate based on $\lambda(z, \xi; \theta)$ and $\hat{\delta}$ is a consistent estimate of δ (e.g., the OLS estimate). Define $\Sigma = \text{avar}(\hat{\theta} - \hat{\delta}\alpha) = \text{avar}(u)$, that is,

$$\Sigma = \text{avar}(\hat{\theta}) + \text{avar}(\hat{\delta}\alpha) - \text{avar}(\hat{\theta}, \hat{\delta}\alpha) - \text{avar}(\hat{\theta}, \hat{\delta}\alpha)'$$

With consistent estimates of the above $\text{avar}(\cdot)$ matrixes in hand, a consistent estimate of Σ , $\hat{\Sigma}$, can be formed and a feasible MD estimator can be obtained as

$$\hat{\alpha}_{\text{MD}} = (\hat{\delta}'\hat{\Sigma}^{-1}\hat{\delta})^{-1}\hat{\delta}'\hat{\Sigma}^{-1}\hat{\theta}$$

whose asymptotic covariance can be estimated by $(\hat{\delta}'\hat{\Sigma}^{-1}\hat{\delta})^{-1}$ (Newey (1985)). Consistent estimates of the $\text{avar}(\cdot)$ matrixes can be obtained in the standard “sandwich” manner by using the cross products of the first-order expansions of $\sqrt{N}(\hat{\theta} - \theta)$ and $\sqrt{N}[(\hat{\delta} - \delta)\alpha]$.

In many applications researchers may be more willing to maintain an assumption on the reduced-form structure for x than to maintain assumptions on the parametric forms for the conditional distributions of unobservables, $h(\eta|x; \gamma)$, and outcomes, $f(y|x, \eta; \alpha)$. If so, then this 2SQML approach could be useful in applications. The fact remains, however, that confidence about the structure of reduced forms is often low, suggesting the likely general utility of the IV-T method, whose consistency relies on less stringent assumptions than needed for consistency of the alternatives considered in this section.²¹

²¹ Indeed, Monte Carlo simulations (available from the author on request) suggest that misspecification of the reduced form for x can have dire implications for the 2SQML method. These simulations also indicate how applications of the IV-A method can result in seriously biased estimates as well.

V. Two Econometric Applications: Cigarette Demand Functions and Birthweight Production Functions

Among the vexing issues involved in the econometric analysis of data containing zero or corner solution outcomes is how to handle estimation when observed regressors may be correlated with unobservables. In a Tobit framework, Newey (1985) provides one of the most distribution-free, computationally feasible solutions to such problems, but even his suggested method requires specification of a parametric reduced form for the endogenous regressors and a conditional symmetry assumption on the unobservables. Jorgenson (1961) was perhaps the first to consider the application of Poisson-type regression models to consumer demand data containing corner solution outcomes. In the spirit of Jorgenson's suggestion, application of the IV-T method to the analysis of such data is perhaps worth considering. Two applications are presented in this section.²²

The first application is to a model of cigarette smoking. Mullahy (1985) suggested that a static structural demand function for cigarettes would depend, in part, on the “stock” of smoking habits K amassed over an individual's lifetime. As such, it has the character of a lagged endogenous variable used as a regressor since $\text{cov}(K, \Theta)$ is thus unlikely to be zero since today's unobserved determinants of smoking behavior are likely to be correlated with yesterday's, last year's, and last decade's. While Mullahy (1985) used simultaneous-equation Tobit-type estimators, the demand model might also be specified as an exponential regression model to which the IV-T method might be applied.²³

The data set for this analysis consists of 6160 observations on males in 1979 and early 1980 from the Smoking Supplement to the 1979 National Health Interview Survey (see Mullahy (1985) for details). The dependent variable is the number of cigarettes currently smoked per day, measured in packs (i.e., the number of cigarettes smoked per day divided by 20). The regressors x are: age in years (*AGE*) and its square; schooling in years (*EDUCATION*) and its square; number of family members (*FAMILY SIZE*); race (*WHITE*, dummy variable); family income in thousands (*INCOME*); the state-level average per-pack price of cigarettes in 1979 (*CIGARETTE PRICE*); an indicator of whether the individual's state of residence had restrictions on smoking in restaurants in place in 1979 (*RESTAURANT RESTRICTION*, dummy variable); the smoking habit stock measure K (*HABIT STOCK*); and a constant. The instrumental variables used include all the elements of x except *HABIT STOCK*, plus the following: cubic terms in, and an interaction between, *AGE* and *EDUCATION*; the one-year lag of *CIGARETTE PRICE*; and a measure of the number of years the state's restaurant smoking restrictions had been in place in 1979.

²² The data sets used in both applications reported in this section are available in Gauss, Stata, or ASCII formats on request from the author; the estimation program, written in Gauss, is also available on request. Note that the asymptotic t -statistics reported for the Poisson ML estimates in both examples are based on robust covariance matrix estimators.

²³ See Wasserman et al. (1991) for a related application.

Table 1 displays the Poisson ML, 2SQML, and IV-T estimates of this structural demand model. The IV-T point estimate of the *HABIT STOCK* parameter is only slightly greater than one-half of the corresponding Poisson ML and 2SQML point estimates. The result vis-à-vis ML is consistent with prior expectations that a positive correlation between K and unobserved demand determinants (i.e. $\text{cov}(K, \Theta) > 0$) would result in upwardly biased estimates of the true structural $\partial y / \partial K$ effect. The discrepancy between IV-T and 2SQML suggests that a linear reduced-form specification for K may be inappropriate.

In so far as the policy variables are concerned, both the *CIGARETTE PRICE* and the *RESTAURANT RESTRICTION* point estimates are negative, as expected, with the IV-T estimate of price elasticity slightly smaller than the ML estimate (-0.54 versus -0.57 at the sample mean of price), and the IV-T point estimate of the *RESTAURANT RESTRICTION* effect considerably larger than the corresponding ML result. Note too that the magnitude of the estimated *INCOME* effect is much larger for IV-T than for the other estimators. Finally, note that the value of the J -statistic is tolerably small.

The second application involves estimation of a model of birthweight determinants. The basic idea here follows that

TABLE 1.—CIGARETTE DEMAND FUNCTION ESTIMATES

Variable (1)	Estimation Method		
	ML (2)	2SQML ^a (3)	IV-T (4)
<i>HABIT STOCK</i>	0.0055 (24.8)	0.0055 (1.5)	0.0032 (1.4)
<i>CIGARETTE PRICE</i>	-0.0094 (2.8)	-0.0051 (0.9)	-0.0089 (2.0)
<i>RESTAURANT RESTRICTION</i>	-0.047 (0.8)	-0.075 (0.9)	-0.062 (1.2)
<i>INCOME</i>	-0.0028 (0.9)	-0.0015 (0.4)	-0.0065 (2.3)
<i>AGE</i>	0.0087 (1.0)	0.054 (0.9)	0.093 (2.1)
<i>AGE</i> ²	-0.00031 (3.2)	-0.00074 (1.3)	-0.0012 (2.8)
<i>EDUCATION</i>	0.035 (1.4)	0.077 (1.3)	0.14 (4.6)
<i>EDUCATION</i> ²	-0.0031 (2.7)	-0.0050 (1.5)	-0.0093 (7.1)
<i>FAMILY SIZE</i>	-0.0081 (0.7)	-0.00058 (0.03)	-0.012 (1.0)
<i>WHITE</i>	-0.051 (1.0)	-0.028 (0.3)	-0.090 (1.3)
Constant	-0.84 (2.9)	-2.07 (2.3)	-2.58 (4.2)

GMM J -test statistic, $\Gamma_N(\hat{\alpha}_{\text{GMM}}) = 7.47$ ($p > .11$; d.f. = 4)

Notes: Dependent variable—cigarettes smoked per day (in packs); number of observations 6160. Asymptotic t -statistics are in parentheses.

^a The 2SQML estimates are obtained by fitting a linear reduced-form model for *HABIT STOCK* by OLS and using the predicted value of *HABIT STOCK* as the included regressor. The t -statistic estimates presented here are not corrected for the inclusion of a predicted regressor.

TABLE 2.—BIRTHWEIGHT PRODUCTION FUNCTION ESTIMATES

Variable (1)	Estimation Method			
	ML y (2)	2SQML ^a y (3)	IV-T y (4)	2SIV $\ln y$ (5)
<i>CIGARETTES</i>	-0.0046 (5.6)	-0.010 (3.4)	-0.0099 (3.5)	-0.011 (3.0)
<i>PARITY</i>	0.014 (2.9)	0.017 (3.3)	0.018 (3.3)	0.019 (3.4)
<i>WHITE</i>	0.056 (4.7)	0.054 (4.6)	0.054 (4.4)	0.063 (4.5)
<i>MALE</i>	0.026 (2.9)	0.026 (2.9)	0.027 (2.9)	0.027 (2.6)
<i>CONSTANT</i>	1.93 (132.3)	1.94 (127.0)	1.94 (121.5)	1.91 (105.2)

GMM J -test statistic, $\Gamma_N(\hat{\alpha}_{\text{GMM}}) = 3.85$ ($p > 0.27$; d.f. = 3)

Notes: Dependent variable—birthweight (in pounds); number of observations 1388. Asymptotic t -statistics are in parentheses.

^a The 2SQML estimates are obtained by fitting a linear reduced-form model for *CIGARETTES* by OLS and using the predicted value of *CIGARETTES* as the included regressor. The t -statistic estimates presented here are not corrected for the inclusion of a predicted regressor.

advanced by Rosenzweig and Schultz (1983), who have suggested that some key physiological determinants of birthweights are likely to be correlated with unobservable birthweight determinants.

The data set is a subsample of 1388 observations on birthweights and associated covariates from the Child Health Supplement to the 1988 National Health Interview Survey. The main focus here is on the role of maternal smoking during pregnancy, measured as the typical number of cigarettes smoked per day during pregnancy (*CIGARETTES*). The other explanatory variables, assumed here to be orthogonal to unobservables, are birth order (*PARITY*), race (*WHITE*, dummy variable), child's sex (*MALE*, dummy variable), and a constant. The vector of instrumental variables consists of a constant, *PARITY*, *WHITE*, *MALE*, and covariates that are assumed to affect maternal smoking behavior but have no direct influence on birthweight: maternal schooling, paternal schooling, family income, and the per-pack state excise tax on cigarettes. Since the dependent variable, birthweight (in pounds), is positive for all observations, the relationships discussed in section III between the nonlinear IV-T estimator and the standard log-linear IV estimator can be explored in the context of this application.

The results are displayed in table 2. Comparing columns 2 and 4, it is seen that the estimated impact of maternal smoking on birthweight outcomes is about twice as large for IV-T as for Poisson ML. Unlike the previous example, it is interesting that the 2SQML (column 3) and IV-T estimates are quite close, suggesting that a linear reduced form for *CIGARETTES* may not be unreasonable.²⁴ Finally, the small

²⁴ It is noteworthy that the IV-T estimate of the elasticity of birthweight with respect to cigarettes smoked at the sample mean, -0.021 , is strikingly similar to the estimate of -0.026 obtained by Rosenzweig and Schultz (1983) in their 2SLS Cobb–Douglas specification.

value of the J -test statistic suggests that the instruments may be reasonable ones.

Column 5 of table 2 presents the linear GMM results based on a log transform of the dependent variable. This estimator is the optimal GMM estimator for the log-linear model, sometimes called the two-stage instrumental-variable (2SIV) estimator (White (1982)). Recall from section III that if the model is log-linear in x and the unobservables with ω and z independent, this estimator as well as IV-T should be consistent. Indeed, comparison of columns 4 and 5 reveals remarkable similarity of the IV-T and the 2SIV point estimates and standard error estimates. While hardly a formal test of model specification, it is nonetheless interesting to assess the relative performance of these two approaches in the context of a single example.

VI. Summary

This paper has proposed an approach to IV estimation of count data or other exponential regression models in which unobserved heterogeneity is correlated with measured regressors. It was suggested that the key orthogonality conditions required for the consistency of GMM estimators will occur only under very special circumstances when the residual function is specified, as might seem natural, as $u_A(y, x; \alpha) = y - \exp(x\alpha)$. The IV-T approach, which instead specifies the residual function as $u_T(y, x; \alpha) = T(y, x; \alpha) - 1$, provides a general solution to this problem.

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